

ECE 20007 FINAL PROJECT: AUDIO EQUALIZER

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Objective

The objective of this project was to utilize concepts of analog signal processing, specifically filters and amplifiers, to design and fabricate a functional audio equalizer. The primary goal was to create a circuit capable of accepting an audio source (such as a phone or computer) and splitting the frequencies, allowing the user to manually adjust the signal gain for specific frequency bands: low (bass), middle, and high (treble). Additionally, the project aimed to amplify the processed signal using a power amplifier to drive a speaker output.

Key functions

The key functions of the project are:

- Filtering: Splitting the input signals into three bands (low, mid, high)
- Volume control: Control the amplitude of each frequency after splitting
- Filtering: filter the signals for noise before recombination
- Signal recombination: Recombine all 3 signals
- Power amplification: Increase the power of the signal enough to drive speakers

Applications of audio equalizers

Audio equalizers serve multiple roles across multiple industries, ranging from professional music production.

- Music Production and Audio Engineering: In recording studios, equalizers are the primary tools used for mixing and mastering. Producers use them to shape the tonal

balance of individual instruments, ensuring they do not "mask" or compete with one another in the frequency spectrum.

- Consumer Electronics (Car and Home Audio): In consumer devices, equalizers allow users to tailor sound to their personal preference or to correct for the physical limitations of the listening environment.

Audio equalizers are an essential part of many people's audio setups. However, most modern implementations use digital algorithms instead of analog equivalents. This project aims to create an analog version of an audio amplifier.

Theory

Filter working principles

Filters are circuits that allow certain frequencies to pass while not allowing others. This project will use three types of filters

For a first order RC filter, the cutoff frequency is

$$f_c = \frac{1}{2\pi RC}$$

- Low pass filters: Allow frequencies below a certain cutoff frequency pass

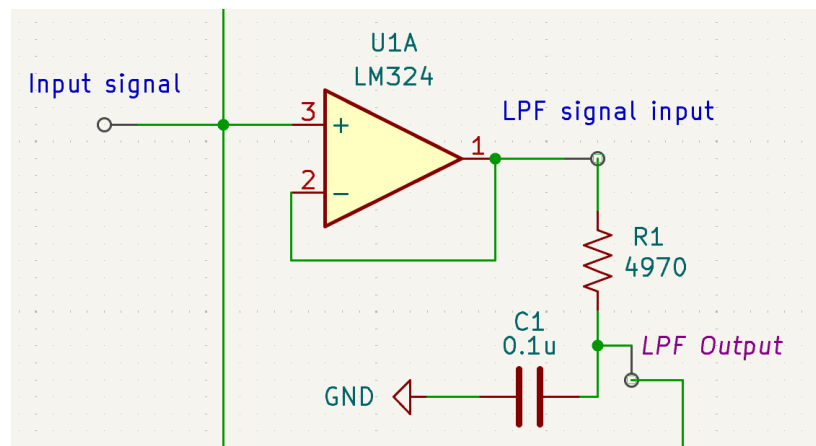


Figure 1: Low pass filter, as connected in final circuit

- High pass filters: Allow frequencies above a certain cutoff frequency pass

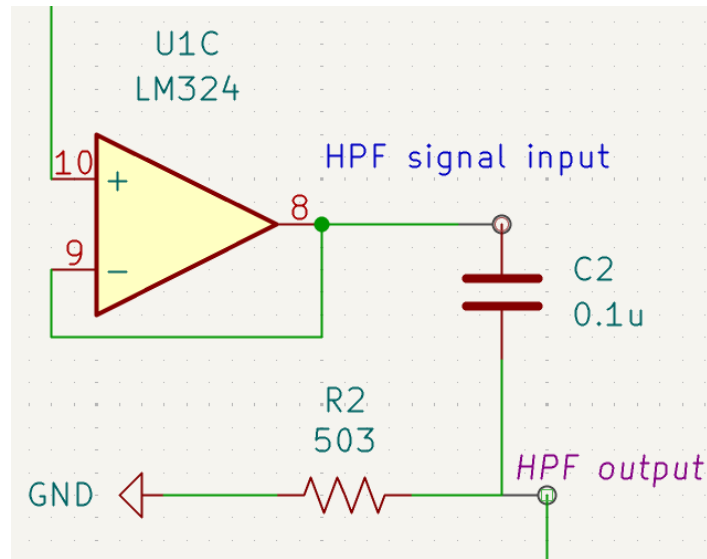


Figure 2: High pass filter, as connected in final circuit

- Band pass filters: Allow a specific range of frequencies to pass

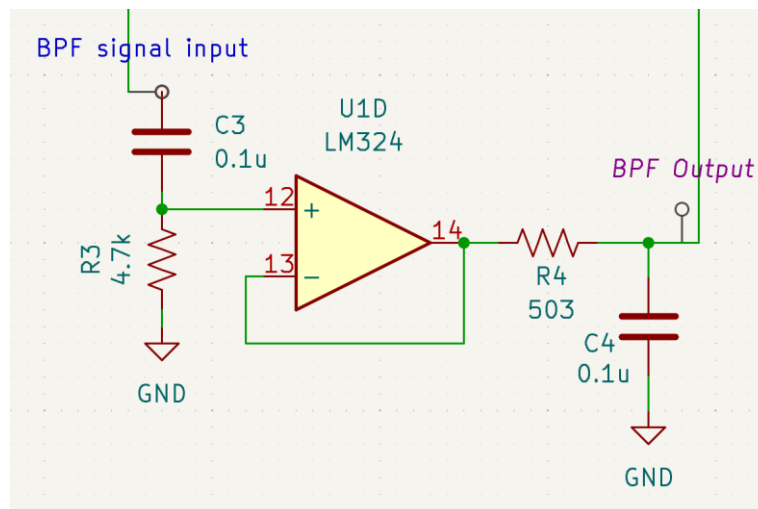


Figure 3: Band pass filter, as connected in final circuit

Operational Amplifier working principle

The op-amps are used here for buffering, filtering (active filters), and summation of waves.

- Buffering: The op amps are used as buffers to prevent components down the line from affecting the resistance at the capacitor and changing the cutoff frequency, effectively isolating the signal.

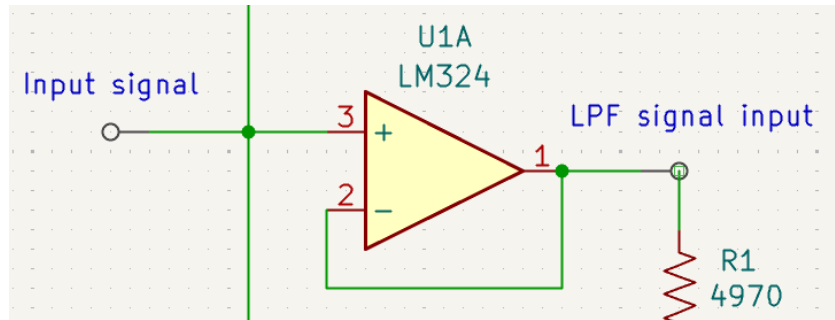


Figure 4: Buffering circuit as applied in the low pass filter

- Summing amplifier: By using the inverting summing amplifier configuration, multiple signals can be combined without significantly interfering with each other

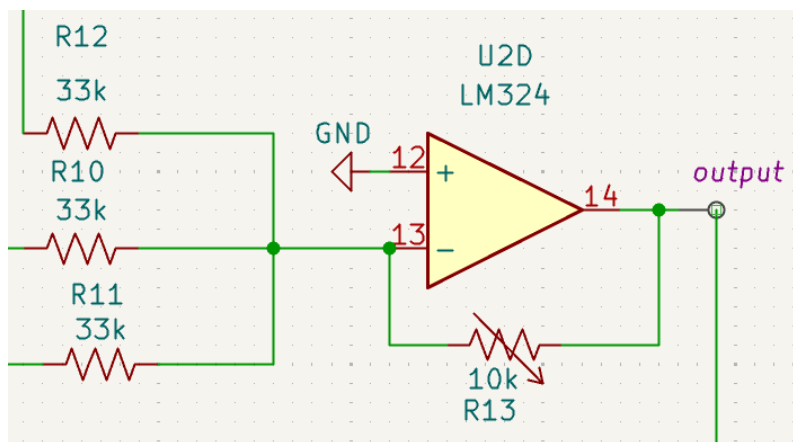


Figure 5: Combination circuit as applied in the low pass filter

Power Amplifier working principle

While the LM324 works well for voltage amplification, it is not enough to drive powerful speakers, as we need >400mW of power. As a replacement, we use the LM386 audio amplifier as a replacement for the LM324 as it is designed for higher current outputs.

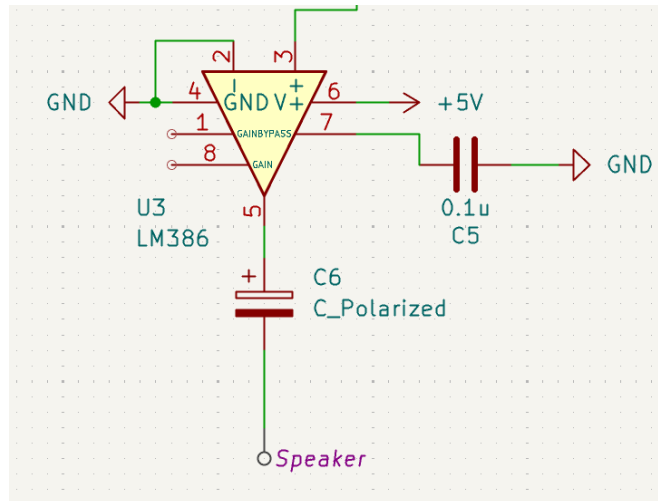


Figure 6: Gain circuit using the LM386

Procedure

Full schematic:

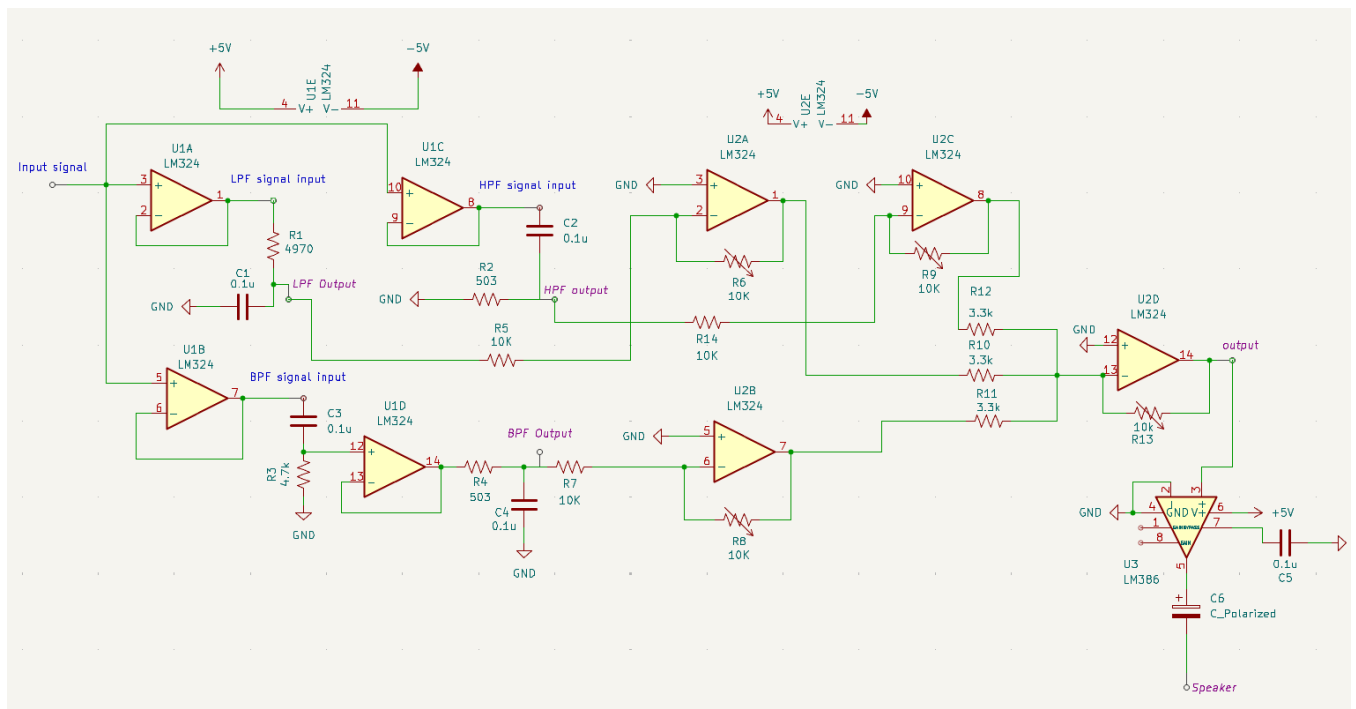


Figure 7: Full circuit schematic

Design parameters and Calculations:

Based on the project requirements

Low pass filter – target $f_c = 320Hz$

Selected value of C = 0.1 μ F

$$f_c = \frac{1}{2\pi RC}$$

$$R = 4970\Omega$$

Closest resistance in kit = 4.7k Ω + 220 Ω

High pass filter – target $f_c = 320Hz$

Selected value of C = 0.1 μ F

$$f_c = \frac{1}{2\pi RC}$$

$$R = 497\Omega$$

Closest resistance in kit = 470 Ω + 33 Ω = 503 Ω

Band pass filter – requirements target *between* $f_c = 320Hz$ and $f_c = 3.2kHz$

Selected value of C = 0.1 μ F

$$f_c = \frac{1}{2\pi RC}$$

$$R1 = 497\Omega \text{ and } R2 = 4970\Omega$$

To buffer and split the signal into 3, we use a LM324 IC, which comes with 4 inbuilt comparators.

Volume control

Volume control was implemented by using an adjustable gain amplifier using an op amp with a 10kΩ potentiometer

$$\text{Max output } V = 100\text{mVrms}$$

$$\text{Min output } V < 15\text{mVrms}$$

$$\text{Total gain } (G) = \frac{V_{out}}{V_{in}} = \frac{100}{320} = 0.3125$$

$$\text{Volume gain } (G_{vol}) = -\frac{R2}{R1} = \frac{10k}{10k} = -1$$

Recombination

Since the total gain should not exceed 0.3125 as calculated before, so

$$G_{total} = G_{vol} * G_{sum}$$

$$G_{sum} = -\frac{Ri}{Rf}$$

$$Rf = 10k\Omega$$

$$R_{in} = -0.3125 * 10k = 3125\Omega$$

Operation description

The input audio signal (320mV RMS from the aux input) is split into three parallel branches. Each branch passes through a specific filter (Low, Mid, or High) to isolate its frequency band. These filtered signals are passed through potentiometers which adjust the amplitude based on user input (volume control). The adjusted signals meet at a summing amplifier, which recombines them into a single waveform. Finally, this signal is fed into the LM386 power amplifier to drive the 8-ohm speaker load.

Results

Low pass FRA

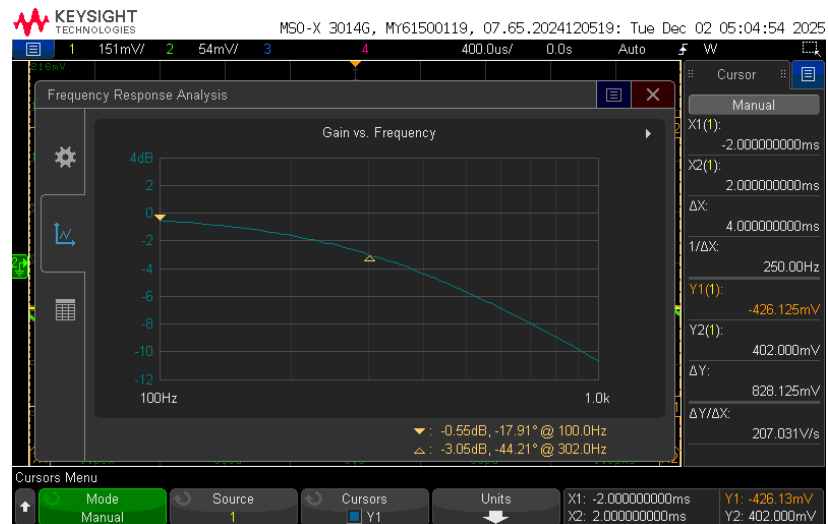


Figure 8: FRA of low pass filter

As observed, the -3dB cutoff frequency is 302Hz (5.6% error from expected value of 320Hz)

This means that frequencies below 320Hz are let through by the filter

Band pass FRA

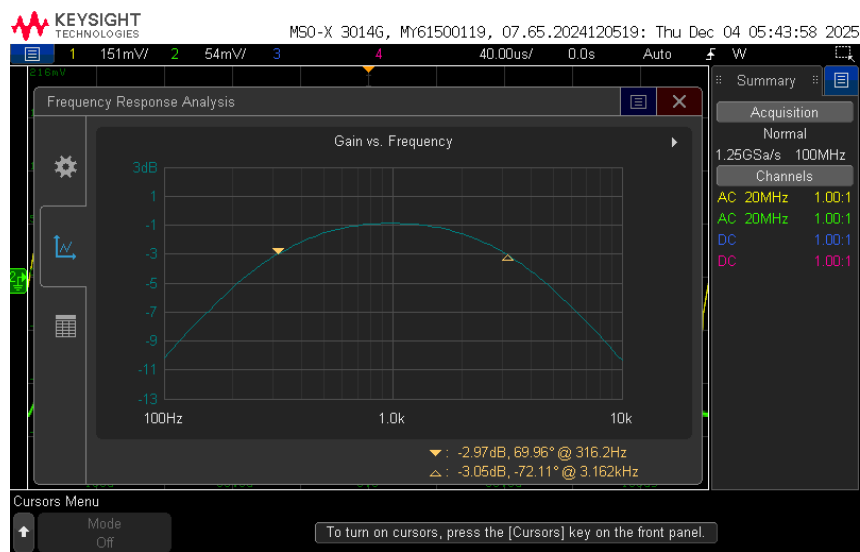


Figure 9: FRA of band pass filter

As observed, the -3dB cutoff frequencies are 316Hz (1.25% error from expected value 320Hz) on the low end and 3.162kHz (1.18% error from expected value of 3.2kHz) on the high end. This means that frequencies between 320Hz and 3.2kHz are let through by the filter

High pass FRA

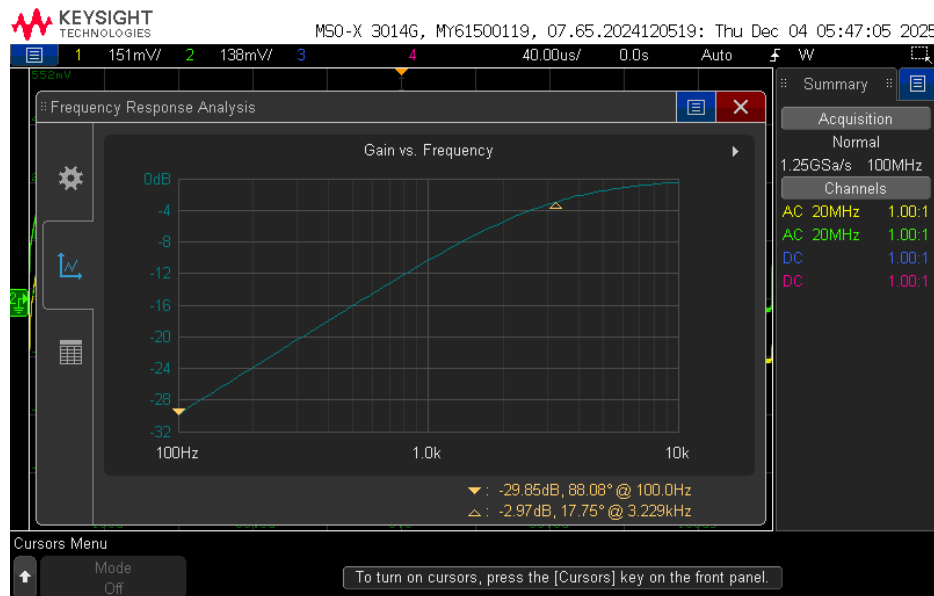


Figure 10: FRA of high pass filter

As observed, the -3dB cutoff frequency is 3.2kHz (0.9% error from expected value of 3.2kHz). This means that frequencies above 3.2kHz are let through by the filter

Vamp (Minimum Volume Settings)

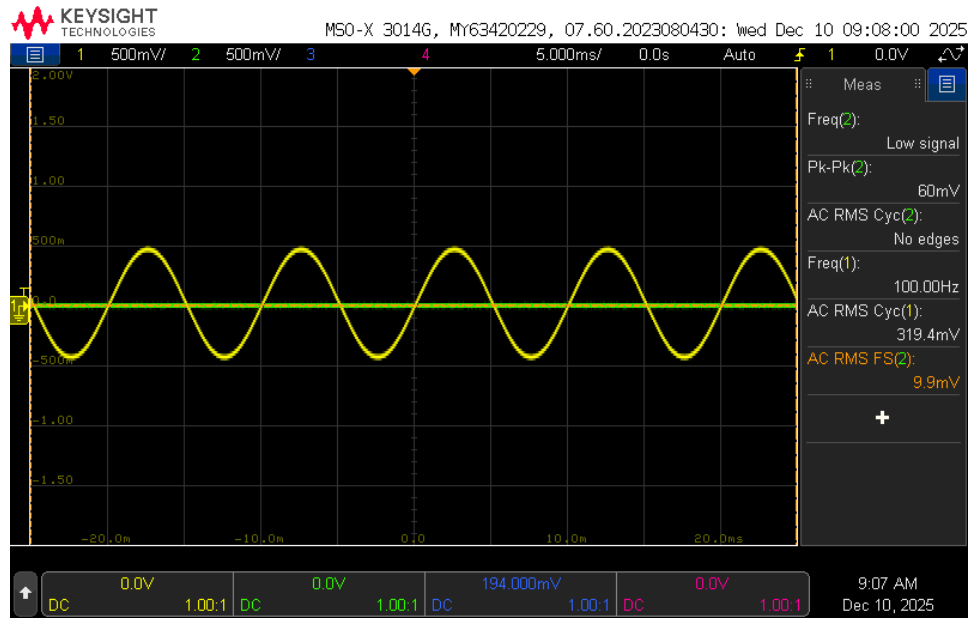


Figure 11: Oscilloscope output at 100Hz

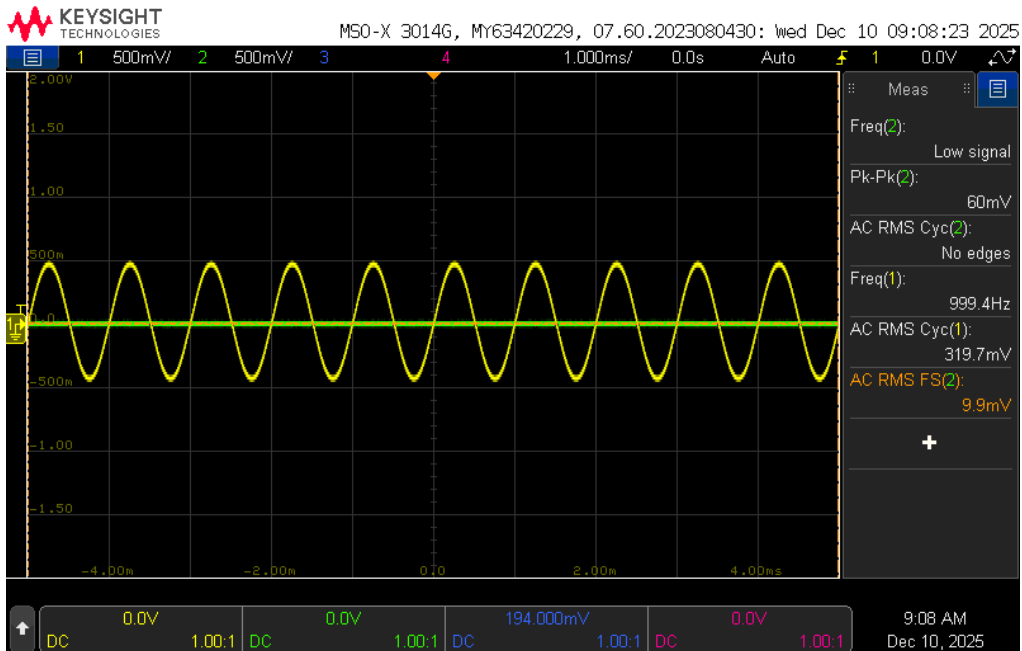


Figure 12: Oscilloscope output at 1kHz

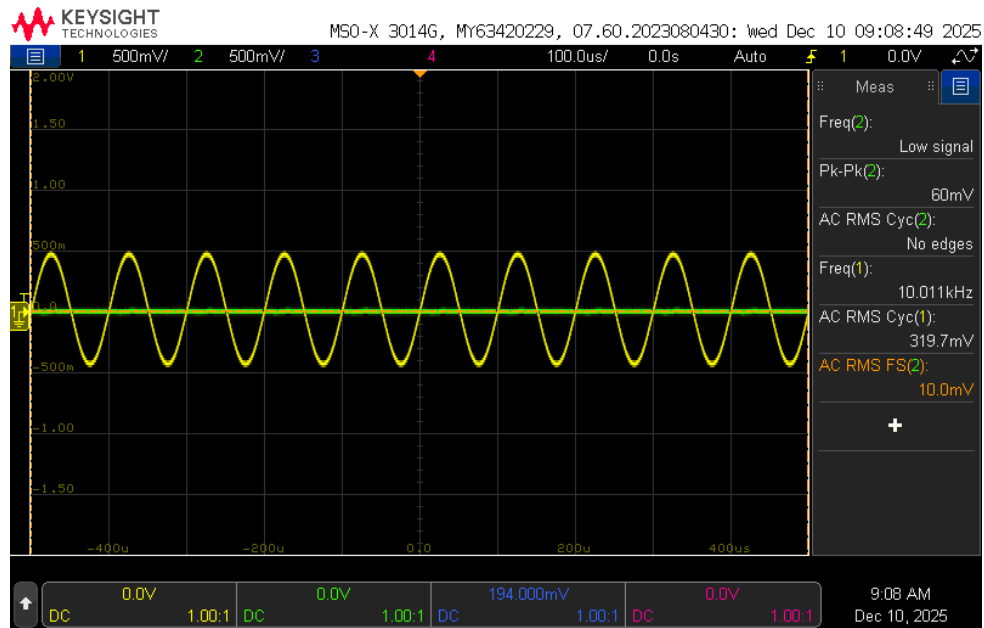


Figure 13: Oscilloscope output at 10kHz

As observed, all of the AC VRMS values are well below the 15mV leakage threshold, which means they are getting suppressed at minimum volume settings

Vamp (Maximum Volume Settings)

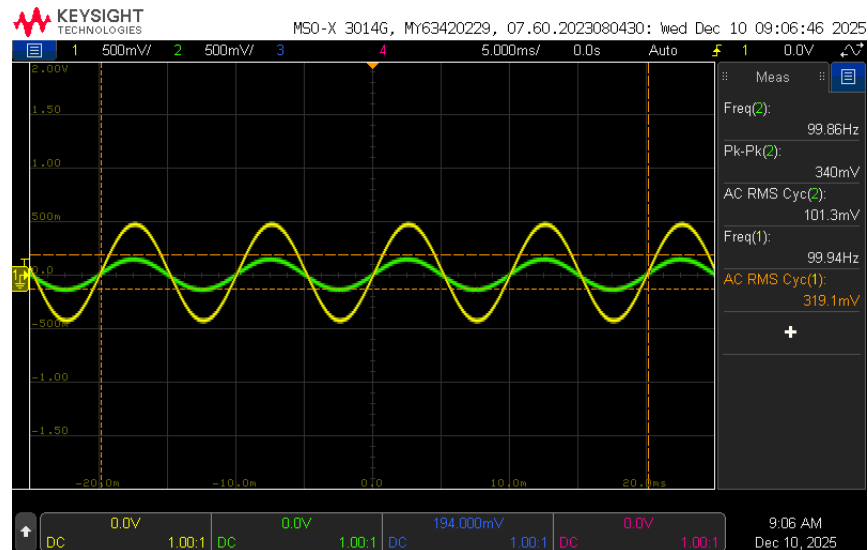


Figure 14: Oscilloscope output at 100Hz

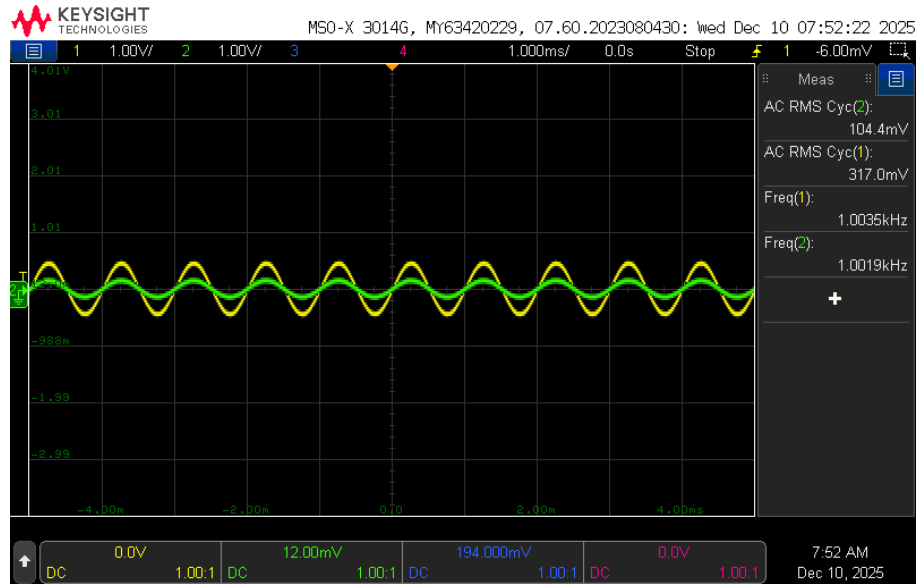


Figure 15: Oscilloscope output at 1kHz

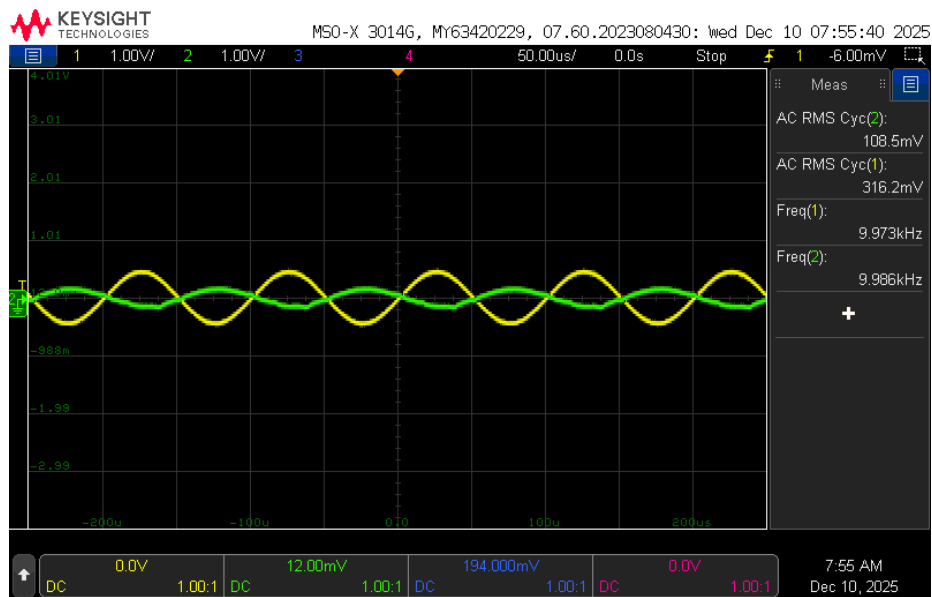


Figure 16: Oscilloscope output at 10kHz

The target was $100\text{mVRMS} \pm 10\%$, which all of our values meet, successfully demonstrating volume control.

Ripple calculation

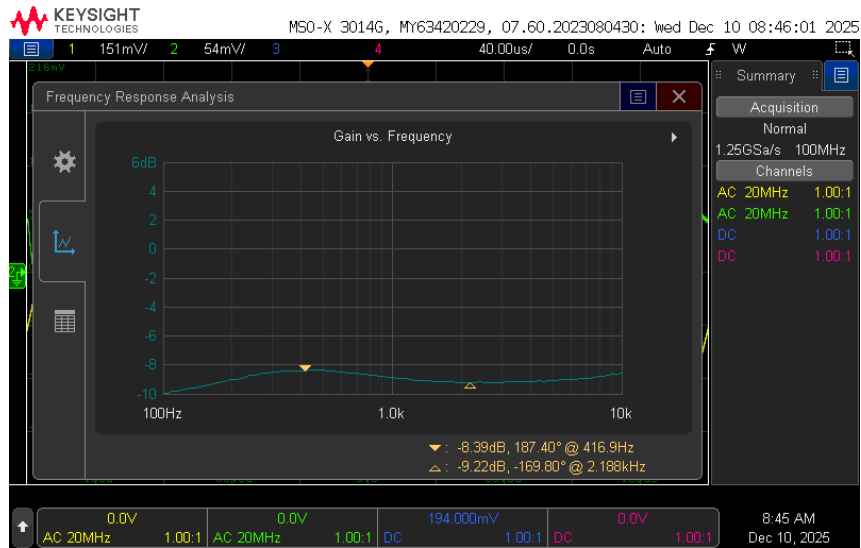


Figure 17: FRA showing ripple frequencies

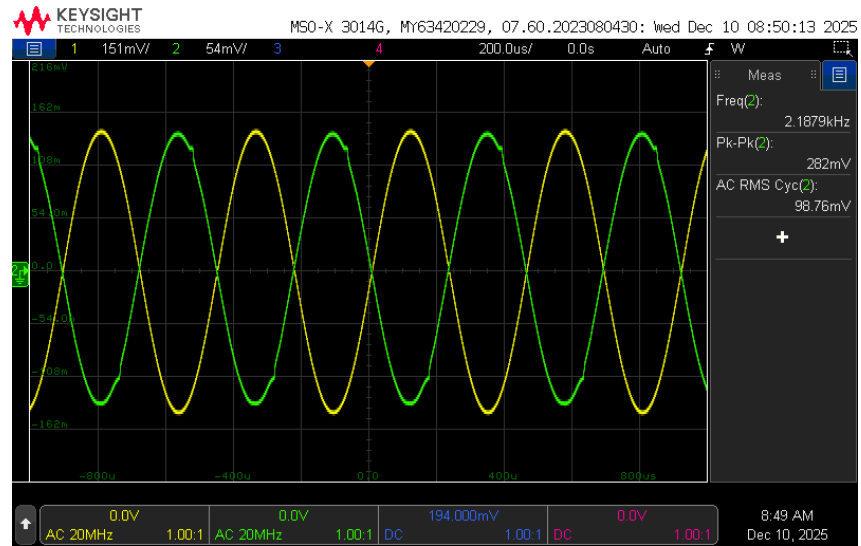


Figure 18: Oscilloscope measurement for minimum ripple

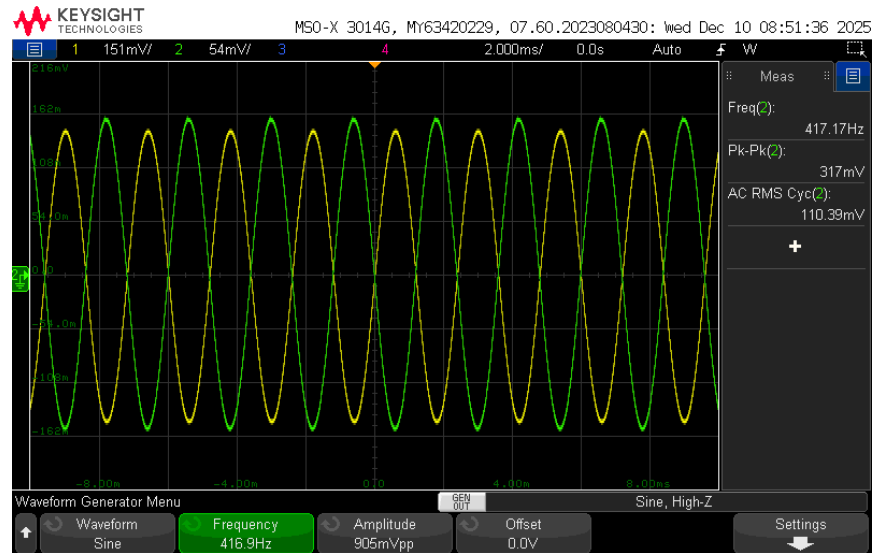


Figure 19: Oscilloscope measurement for maximum ripple

$$\text{Ripple} = V_{\max} - V_{\min} = 110.39\text{mV} - 98.76\text{mV} = 11.63\text{mV}$$

This meets our specification of 15mV RMS

Power Amplifier output

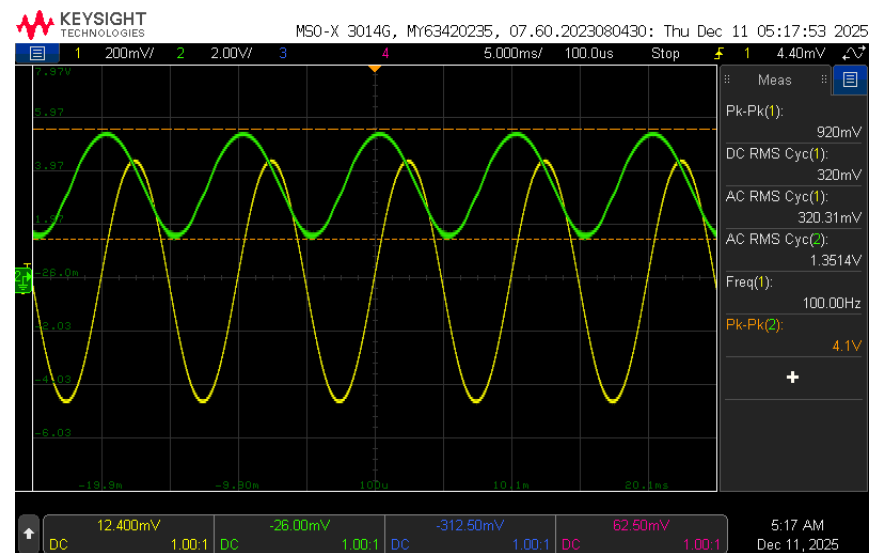


Figure 20: Oscilloscope output at speaker input

Using the measured VRMS across the 8Ω speaker:

$$V_{RMS} = \frac{V_{Pp}}{\sqrt{2}} = \frac{4.1}{\sqrt{2}} = 2.899V$$

$$P = \frac{V_{RMS}^2}{R_{speaker}} = \frac{1.78^2}{8} = 1050mW$$

This is more than enough than the minimum requirement of 400mW of power

Results:

All of the values calculated and observed are within the 10% experimental range of errors. Most errors were most likely because of the resistances not being exact the same as the calculated and tolerances of the resistors and the capacitors.

Conclusion

In this project, we successfully designed and implemented a three-band audio equalizer with power amplification. The circuit successfully filtered frequencies into bass, mid, and treble bands and allowed for volume control for all the different frequencies. The LM386 amplifier successfully drove the speaker.

No errors were too large beyond the expected experimental errors, caused by tolerances of the resistances and capacitors

References

1. ECE20007 Final Project: Audio Equalizer, Purdue University, Fall 2025.
2. "Inverting Operational Amplifier," *Electronics Tutorials*, Available: https://www.electronics-tutorials.ws/opamp/opamp_2.html.
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